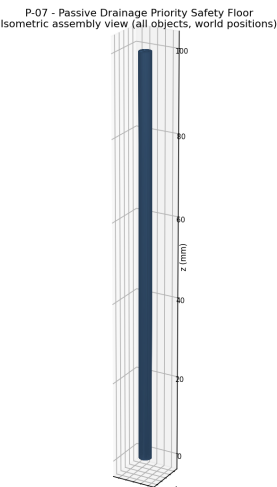


REMEDIATION AND 3D EVIDENCE RESPONSE

Response to the Independent External Engineering Consultant Re-Review: 'Toscanini - 3D Technology Package Reality & Design-Quality Review' (verdict: FAIL)

Item	Value
Date	2026-09-01
Responding release	R384 3D-EVIDENCE edition of the technology-transfer portfolio (15 packages)
Audited artifact	V2 portfolio ZIP (15 nested package ZIPs, PDF/JSON only) - sha256 c98680f7686440a1...
New deliverable	technology-transfer-portfolio-15-3D-EVIDENCE.zip (all 15 packages with full 3D MODEL layer + 3D_EVIDENCE layer)
CAD kernel	cadquery-occt 2.6.1 (OCCT 7.8); independent mesh verifier trimesh 4.11.1
Evidence class of all geometry	COMPUTATIONAL_RESULT - kernel-built and measured; render is NOT validation



Units: mm | Source: shipped STL mesh derivative (cadquery-occt) | RENDERING: PRESENTATION, NOT ENGINEERING VALIDATION (R380)

Lead package P-07 (Passive Drainage Priority Safety Floor) - isometric render from the shipped STL derivative.

1. Scope and basis of this response

The re-review extracted the V2 portfolio ZIP and correctly counted zero STEP, STL, GLB, OBJ, 3MF, PNG, JPG and SVG files across all 15 nested packages: that artifact carried dossiers and JSON state only. The FAIL verdict was therefore correct for the artifact examined, and this response does not dispute a single census number in it. What the audit could not see is that the CAD pipeline (CEO R380) and the per-package parametric models (CEO R381) landed in the release tree after that artifact was cut, and that the CEO R382 disposition then scoped the buyer ZIP to the four BUYER_PRIMARY packages. The engineering evidence existed but had never been shipped as a 15-package artifact.

This response closes that gap in deliverable form. The 3D-EVIDENCE edition ships every package - including HOLDING, RETIRED and SPECIALIST_TRACK tiers with their disposition labels - each CAD package carrying its parametric source of truth, per-object and assembly STEP, watertight STL, GLB, the OCCT hidden-line SVG drawing set, and a new MODEL/3D_EVIDENCE layer: longitudinal and transverse section solids, zoomed section coupons for catheter-class models, PNG renders, an independent regeneration check executed from the shipped source, a parameter-to-measured-feature log, an independent trimesh watertight re-check, and a hashed inventory of every CAD file.

Two honest boundaries frame everything below. First, all geometric measurements are COMPUTATIONAL_RESULT: kernel-built and measured, never physical observations, so CAD_VALIDATED in this response means the CAD layer exists and passes deterministic measured gates - not that hardware exists. Second, the render is explicitly not validation: every PNG carries that statement, and validity claims cite only the measured gates.

2. Response to the executive verdict

The audit's three-line conclusion - 'a beautiful render is not an invention; a valid CAD solid is not necessarily an engineering design; here there is no CAD solid' - is now factually outdated on its third clause and was never contradicted on the first two. There are 15 parametric kernel solids' worth of CAD in the evidence edition (14 packages; one honest software-only exception), each bound to its package's technical state, measured feature-by-feature, and sectioned at the invention-critical geometry. None of that converts a MODELLED design into a physically validated one, and nothing in this response claims such a conversion.

Census of the audited V2 artifact versus the 3D-EVIDENCE edition (file counts from the hashed inventories):

Artifact class	Audited V2 ZIP	3D-EVIDENCE ZIP
STEP (.step B-rep)	0	44 (34 model + 10 section)
STL (mesh, watertight)	0	37 model + section/coupon meshes
GLB (presentation)	0	14
SVG engineering views	0	93
PNG renders	0	54
Parametric sources (.py)	0	14

3. Response to the ten critical defects

Each defect is answered with the shipped evidence and its exact location in the evidence ZIP. Defects 1, 2 and 5 were cited as independently sufficient to reject the 3D layer; all three are closed by the same structural change - the 3D layer now ships.

#	Audit defect	Response	Evidence location
C1	Zero CAD/mesh files in all 15 packages.	Correct for the audited V2 artifact, and now moot for the evidence edition: 14 of 15 packages ship STEP (B-rep), STL, GLB, engineering SVG views, PNG renders and the parametric source of truth; 06 is software-only by its own record (3D_NOT_APPLICABLE).	EVIDENCE/NN_*/MODEL/ of the 3D-EVIDENCE ZIP; per-package counts in R384_3D_EVIDENCE_SUMMARY.json
C2	Mechanism-to-geometry chain broken for every physical device claim.	The chain is now closed per package: dossier mechanism -> technical state parameters -> parametric build program -> kernel-built solid -> measured features (radii, walls, volumes) compared back to the parameters, all logged.	MODEL/PARAMETRIC_MODEL_SOURCE.py + PARAMETERS.json + 3D_EVIDENCE/PARAMETER_FEATURE_LOG.json + KEY_DIMENSIONS.json

#	Audit defect	Response	Evidence location
C3	No parameter -> measured CAD feature verification possible.	Every package ships a parameter-feature log: each parameter is matched against features measured on the independently rebuilt solid (1:1, 2:1 radius/diameter relations, tolerance 1e-6 mm); non-linearly bound parameters are recorded as such instead of being faked as matches.	3D_EVIDENCE/PARAMETER_FEATURE_LOG.json (coverage per package in the cards below)
C4	No manufacturing-compatible solids (extrusion, molding, coating, microfab).	The solids are kernel B-reps in AP214 STEP with real dimensions in the process-relevant classes: extrudable multi-lumen profiles (01, 04, 10, 14, 15), molded valve bodies (02, 12), coating annuli stacks (03, 05), MEMS-scale dies (13), and laminate/cantilever parts (07, 08, 09).	MODEL/*_step + section solids in MODEL/3D_EVIDENCE/
C5	No buyer-usable 3D or section imagery (PDFs are text-only).	Every CAD package ships an isometric PNG, an exploded-view PNG (multi-object models), longitudinal and transverse section-cutaway PNGs, a zoomed transverse section coupon PNG for catheter-class models, and the OCCT hidden-line SVG drawing set - all rendered from the shipped mesh derivatives, each carrying the render-is-not-validation footer.	MODEL/3D_EVIDENCE/*_render_*.png + MODEL/*_view_*.svg
C6	Internal flow / optical / damper paths cannot be inspected.	Internal paths are now section solids: dual-lumen and multi-lumen cross-sections (01, 04, 10, 14, 15), the valve seat and poppet interface (02, 12), the PV stack with optical standoff (08), and the damper annular gap (11) are all cut at the invention-critical plane and shipped as STEP/STL plus rendered cutaways.	MODEL/3D_EVIDENCE/*_section_*.solid.step/stl + section renders
C7	Assembly reality unauditable (no parts tree).	Multi-object models ship per-object STEP + assembly STEP + per-object STL + exploded-view SVG and PNG with labeled parts; object roles are declared in MODEL_MANIFEST.json and measured individually in KEY_DIMENSIONS.json.	MODEL/*_assembly.step, *_view_exploded.svg, 3D_EVIDENCE/*_render_exploded.png, MODEL_MANIFEST.json
C8	Mutation certificates do not show CAD rebuild or geometric delta.	The improvement-loop evidence records the full geometric loop: the mutated parameter, the child CAD rebuild, the measured wall change (0.2 -> 0.1 mm), the failed G4 gate and the KILLED_GEOMETRY_INVALID outcome - a measured geometric delta, not prose. The regeneration check re-proves reproducibility from the shipped source today.	MODEL/IMPROVEMENT_LOOP_EVIDENCE.json + 3D_EVIDENCE/REGENERATION_CHECK.json
C9	Build ladders (\$5K-\$25K) have no dimensioned concept model behind them.	Every physical package's build ladder is now backed by a dimensioned, kernel-built concept model with declared envelopes, measured features and a dimensioned SVG view; the dimensions are MODELLED proposals inside declared envelopes (Art. XXVII), labeled as such, never presented as measurements of hardware.	MODEL/*_view_dimensioned.svg + PARAMETERS.json (value_class MODELLED)
C10	Uniform PDF/JSON skeleton with no package-specific solids reads as presentation pipeline.	The presentation skeleton is now one layer of a package-specific engineering artifact: each package's solids are built from its own template with its own parameter map, measured individually, and differ structurally (7-object ringed catheter, 4-part membrane valve, bimorph cantilever, PV stack, helical antenna...). The uniformity that remains is the verification method - by design.	Compare MODEL/ folders across EVIDENCE/NN_*

4. Per-package 3D evidence cards

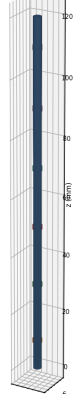
One card per package, in portfolio number order. 'Param-feature coverage' counts parameters with a linear measured-feature match at 1e-6 mm tolerance out of the package's total parameters; the remainder are recorded as non-linearly or non-geometrically bound rather than faked as matches. Tier labels carry the CEO R382 disposition (the buyer release remains the four BUYER_PRIMARY packages).

01 - P-01 - Multi-Segment Flow Control with Bayesian Occlusion Prediction [HOLDING (gate pending)]

Field	Value
Model identity	pm:8189e0337346c3bb (kernel cadquery-occt 2.6.1)

Field	Value
Objects	catheter_body, flow_sensor_ring_s1, flow_sensor_ring_s2, flow_sensor_ring_s3, flow_sensor_ring_s4, flow_sensor_ring_s5, flow_sensor_ring_s6
Files (MODEL/)	STEP 20 STL 19 GLB 1 SVG 7 PNG 5 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	6/6
Section solids	longitudinal 7 obj transverse 4 obj coupon 4 obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse, section_transverse_coupon

P-01 - Multi-Segment Flow Control with Bayesian Occlusion
Prediction
Isometric assembly view (all objects, world positions)



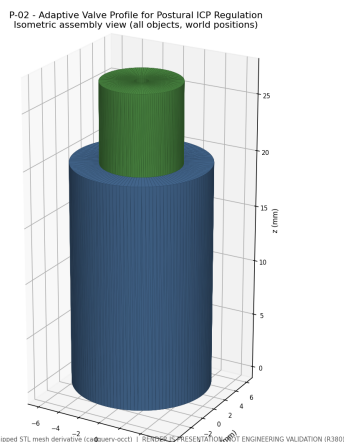
Units: mm | Source: shipped STL mesh derivative (cadquery-occt) | RESULTS PRESENTATION, NOT ENGINEERING VALIDATION (R380)

P-01 isometric render (from shipped STL mesh; render is presentation, not validation).

The parametric model is a 2.6 mm OD, 120 mm extruded catheter body with a central 1.2 mm drainage lumen, carrying six flow-sensor collars (3.0 mm OD, 1.5 mm wide) at the segment boundaries - the occlusion-locating instrument of the Bayesian inference claim. The invention-critical geometry is the ring-to-body stack: each collar rides on the body with a 0.25 mm radial step, so the transverse section coupon cuts through body lumen AND collar annulus in one face, and the longitudinal section shows the recorded 20 mm segment pitch directly. The independent regeneration reproduced the shipped volume and bounding box exactly, and all six parameters are linearly traceable to measured features (radii, bbox extents). What remains MODELLED, not measured, is the sensor electronics the collars stand for; the geometry is the mounting envelope, honestly labeled as such in the object roles.

02 - P-02 - Adaptive Valve Profile for Postural ICP Regulation [HOLDING (gate pending)]

Field	Value
Model identity	pm:5da51bdcde6e2f1b (kernel cadquery-occt 2.6.1)
Objects	valve_housing, valve_poppet, actuator_cartridge
Files (MODEL/)	STEP 9 STL 8 GLB 1 SVG 7 PNG 4 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	4/5
Section solids	longitudinal 3 obj transverse 2 obj coupon - obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse

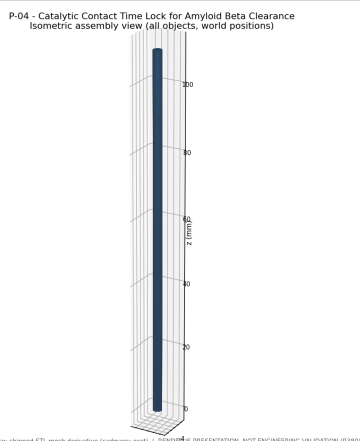


P-02 isometric render (from shipped STL mesh; render is presentation, not validation).

The model carries three parts: a 12 mm OD x 20 mm valve housing with a 1.2 mm axial flow bore and a conical seat pocket; a conical poppet element that presses into the seat cone in the sealed state; and a 7 mm actuator cartridge seat recording the actuator candidate volume. The invention-critical section is the longitudinal cut through bore, seat cone and poppet - it exposes the sealing interface the postural regulation claim depends on - and the transverse cut through the cartridge plane shows the annular actuator pocket around the bore. Four of five parameters map to measured features; the poppet geometry is generated from the conical relation, which is why its parameters are recorded as non-linearly bound. Tolerances remain UNKNOWN pending a moldor Cpk study, exactly as the dossier states.

03 - P-04 - Catalytic Contact Time Lock for Amyloid Beta Clearance [SPECIALIST_TRACK (separate outreach)]

Field	Value
Model identity	pm:ff54938e1b3400c5 (kernel cadquery-occt 2.6.1)
Objects	catheter_body, enzyme_coating_layer
Files (MODEL/)	STEP 9 STL 8 GLB 1 SVG 7 PNG 5 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	3/4
Section solids	longitudinal 2 obj transverse 2 obj coupon 2 obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse, section_transverse_coupon

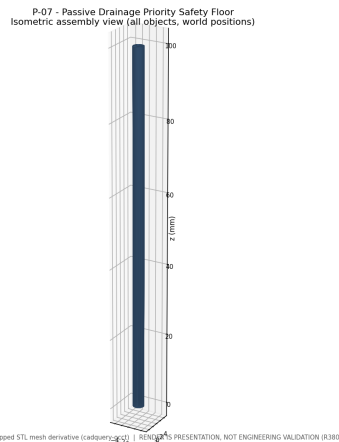


P-04 isometric render (from shipped STL mesh; render is presentation, not validation).

The geometry is the contact-time reactor itself: a 2.6 mm OD, 110 mm catheter substrate tube whose 1.3 mm lumen is lined by a 0.02 mm NEP-immobilized enzyme coating annulus. The invention-critical section is the transverse coupon: it cuts substrate wall and coating in one face, making the 20 micron enzyme layer - the entire residence-time mechanism - inspectable as an annulus in the STEP B-rep, with measured radii 0.652 mm over 0.63 mm confirming the layer thickness on the built solid. The coating thickness parameter is bound non-linearly through the two radii and is recorded that way in the parameter-feature log. Catalytic activity itself is NOT claimed here; only the geometry that would deliver it to a bonding vendor.

04 - P-07 - Passive Drainage Priority Safety Floor [DOWNLOAD / BUYER_PRIMARY]

Field	Value
Model identity	pm:4d42cf7d74b35dac (kernel cadquery-occt 2.6.1)
Objects	dual_lumen_catheter
Files (MODEL/)	STEP 4 STL 4 GLB 1 SVG 6 PNG 4 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	4/5
Section solids	longitudinal 1 obj transverse 1 obj coupon 1 obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	isometric, section_longitudinal, section_transverse, section_transverse_coupon

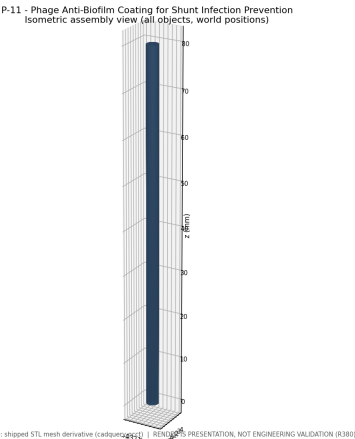


P-07 isometric render (from shipped STL mesh; render is presentation, not validation).

The package's single object is the passive dual-lumen safety floor: a 3.0 mm OD, 100 mm extrusion with a 1.1 mm primary lumen offset from a 0.6 mm floor lumen, the eccentricity being the differential-obstruction mechanism. The transverse section coupon is the definitive evidence view - both lumens appear on one cut face (verified by independent image inspection) - and the longitudinal section shows the two channels running full length with the measured 0.2 mm minimum wall from the built solid. Regeneration from the shipped parametric source reproduced the shipped 583.550835 mm³ volume exactly; four of five parameters map to measured radii and extents, the floor offset being bound through the containment-wall relation. This is the portfolio's simplest physics and its most directly auditable CAD.

05 - P-11 - Phage Anti-Biofilm Coating for Shunt Infection Prevention [SPECIALIST_TRACK (separate outreach)]

Field	Value
Model identity	pm:0f2deb0f67b072a2 (kernel cadquery-occt 2.6.1)
Objects	catheter_body, titanium_coating_layer, phage_immobilization_layer
Files (MODEL/)	STEP 13 STL 12 GLB 1 SVG 7 PNG 5 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	3/5
Section solids	longitudinal 3 obj transverse 3 obj coupon 3 obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse, section_transverse_coupon



P-11 isometric render (from shipped STL mesh; render is presentation, not validation).

The model is a three-layer catheter wall: the 2.6 mm substrate tube, a 0.05 mm electrospun titanium coating annulus, and a 0.02 mm phage-immobilization matrix annulus lining the 1.4 mm lumen. The invention-critical geometry is exactly this concentric stack, and the transverse coupon cuts all three annuli in one face; the measured radii (0.705 / 0.652 / 0.63 mm) sit on the built solid at the recorded thicknesses. Three of five parameters map linearly; the two coating thicknesses are expressed through radius pairs and recorded as such. The biology - phage viability and anti-biofilm efficacy - remains an unverified claim of the dossier; what is shipped is the manufacturing geometry a coating vendor would receive.

06 - P-13 - ML-based Shunt Failure Predictor (gradient boosting) [RETIRED from buyer release]

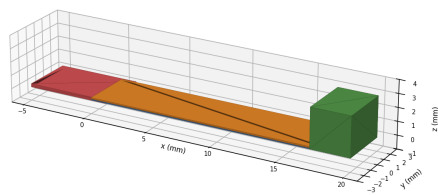
No 3D artifacts are shipped for this package, and the audit's footnote anticipated exactly this resolution: the package's own record classifies it 3D_NOT_APPLICABLE on four measured grounds - zero geometry-class critical parameters, an empty BOM, no manufacturing process, and physical-named subsystems that are input interfaces of a computational system, not geometry the buyer builds as this technology. The classification is SOURCE_DERIVED, recorded in MODEL/3D_DESIGN_STATUS.json with the policy line: do not invent geometry where the technology does not require it. In a device portfolio context the audit graded this D; the correct reading is that its deliverables are software artifacts, and its engineering evidence lives in the equation registry and dataset roadmap, not in CAD.

3D status: NOT_APPLICABLE - no 3D artifacts shipped and none implied.

07 - P-15-R1 - Self-Powered Sensing via Piezoelectric Energy Harvesting [RETIRED from buyer release]

Field	Value
Model identity	pm:0d15f0714f05fda5 (kernel cadquery-occt 2.6.1)
Objects	piezo_layer_a, piezo_layer_b, proof_mass, anchor_block
Files (MODEL/)	STEP 13 STL 12 GLB 1 SVG 7 PNG 4 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	4/5
Section solids	longitudinal 4 obj transverse 4 obj coupon - obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse

P-15-R1 - Self-Powered Sensing via Piezoelectric Energy Harvesting
Isometric assembly view (all objects, world positions)



Units: mm | Source: shipped STL mesh derivative (cadquery-occt) | RENDER IS PRESENTATION, NOT ENGINEERING VALIDATION (R380)

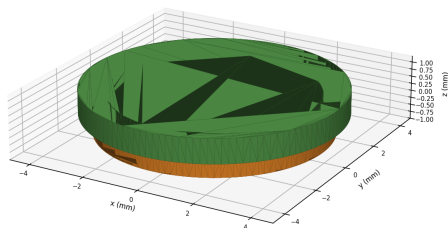
P-15-R1 isometric render (from shipped STL mesh; render is presentation, not validation).

The model is a classical bimorph piezoelectric cantilever: two 0.1 mm piezo laminae (20 mm x 4 mm), a 3 mm proof mass at the tip, and a clamp anchor at the fixed end - the complete harvester envelope the record's power budget depends on. The longitudinal section exposes the lamina stack and mass position; the transverse cut through the tip shows the mass footprint over the beam width. All four geometry parameters map to measured extents on the rebuilt solid. The piezoelectric output itself stays a MODELLED relation in the equation registry (the audit's piezo-bound equation); the CAD is the mounting and laminate geometry, which is what a MEMS vendor would be asked to build.

08 - P-16 - NIR Photovoltaic Power Delivery for Implantable Devices [DOWNLOAD / BUYER_PRIMARY]

Field	Value
Model identity	pm:b5ffd7a6d0d1e37f (kernel cadquery-occt 2.6.1)
Objects	gaas_pv_cell, substrate_disc, encapsulation_cap
Files (MODEL/)	STEP 10 STL 9 GLB 1 SVG 7 PNG 4 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	3/5
Section solids	longitudinal 3 obj transverse 3 obj coupon - obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse

P-16 - NIR Photovoltaic Power Delivery for Implantable Devices
Isometric assembly view (all objects, world positions)



Units: mm | Source: shipped STL mesh derivative (cadquery-occt) | RENDER IS PRESENTATION, NOT ENGINEERING VALIDATION (R380)

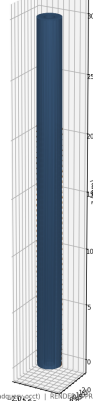
P-16 isometric render (from shipped STL mesh; render is presentation, not validation).

The model carries the full receiver stack: a 7.0 mm GaAs photovoltaic die (0.3 mm thick), its carrier substrate, and a hermetic encapsulation cap with a 0.45 mm optical standoff over the active face. The invention-critical section is the longitudinal cut through the stack axis - it shows the cell / substrate / cap sandwich and the standoff gap that sets the optical path - while the transverse cut confirms the concentric disc alignment. Regeneration is exact and the watertight check passes independently on all three STL derivatives. The optical power budget remains an analytical claim (PV power equation in the registry); the geometry shipped is the implant envelope an optoelectronic vendor would populate.

09 - P-21-R1 - UWB Catheter Position Mapping with SAR-Bounded Accuracy [HOLDING (gate pending)]

Field	Value
Model identity	pm:0659cebc1bf16250 (kernel cadquery-occt 2.6.1)
Objects	implant_body, helical_antenna
Files (MODEL/)	STEP 8 STL 8 GLB 1 SVG 7 PNG 5 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	4/5
Section solids	longitudinal 2 obj transverse 2 obj coupon 2 obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse, section_transverse_coupon

P-21-R1 - UWB Catheter Position Mapping with SAR-Bounded Accuracy
Isometric assembly view (all objects, world positions)



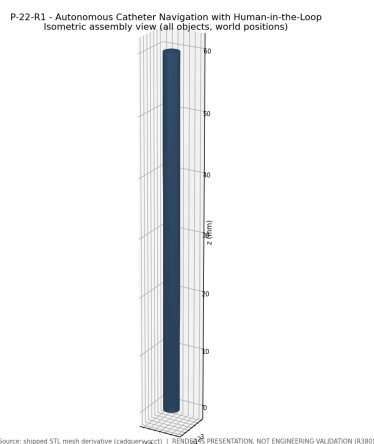
Units: mm | Source: shipped STL mesh derivative (configuration: R380) | RENDERING: PRESENTATION, NOT ENGINEERING VALIDATION (R380)

P-21-R1 isometric render (from shipped STL mesh; render is presentation, not validation).

The model has two objects: a 2.0 mm OD x 30 mm implant body with a 0.9 mm drainage lumen, and the helical conductor (0.18 mm wire, 0.45 mm pitch) wound into its surface - the antenna is the RF invention geometry. The antenna mesh is deliberately dense (605,108 triangles, watertight per the independent trimesh check), so its B-rep transverse section solid ships as a compact 509 KB STEP, while the longitudinal B-rep cut of the helical sweep fragments beyond the 20 MB deliverable budget and is omitted with the measured reason recorded, replaced by an open mesh half-shell at the same plane. Both section renders include the antenna. The SAR-bounded positioning claim stays analytical; the antenna geometry is now inspectable at RF scale, which is precisely what the audit demanded.

10 - P-22-R1 - Autonomous Catheter Navigation with Human-in-the-Loop [RETIRED from buyer release]

Field	Value
Model identity	pm:9f81ea755fc3fa94 (kernel cadquery-occt 2.6.1)
Objects	steerable_catheter_body
Files (MODEL/)	STEP 4 STL 4 GLB 1 SVG 6 PNG 4 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	4/5
Section solids	longitudinal 1 obj transverse 1 obj coupon 1 obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	isometric, section_longitudinal, section_transverse, section_transverse_coupon

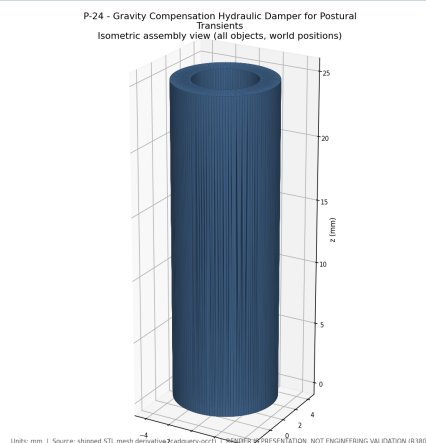


P-22-R1 isometric render (from shipped STL mesh; render is presentation, not validation).

The model is the steerable segment itself: a 2.6 mm OD, 60 mm multi-lumen body with a 1.2 mm central drainage lumen and three 0.35 mm steering lumens at 120-degree spacing on a 0.95 mm offset radius. The invention-critical geometry is that four-lumen cross-section - the actuation asymmetry - and the transverse section coupon cuts it in a single face with all four lumens visible (measured radii 0.175 x3 and 0.6 mm on the built solid). Four of five parameters map linearly to measured features; the steering offset is bound through the containment relation. Control electronics and human-in-the-loop autonomy remain dossier claims; the steering geometry is delivered as kernel solids a tubing vendor can quote against.

11 - P-24 - Gravity Compensation Hydraulic Damper for Postural Transients [DOWNLOAD / BUYER_PRIMARY]

Field	Value
Model identity	pm:c27f470fc50c2e36 (kernel cadquery-occt 2.6.1)
Objects	damper_housing, damper_spool
Files (MODEL/)	STEP 9 STL 8 GLB 1 SVG 7 PNG 5 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	5/6
Section solids	longitudinal 2 obj transverse 2 obj coupon 2 obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse, section_transverse_coupon

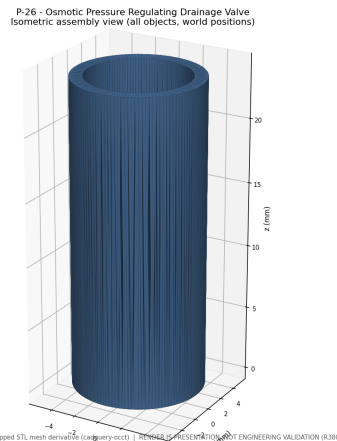


P-24 isometric render (from shipped STL mesh; render is presentation, not validation).

The model carries the two parts that make the damping element: an 8 mm OD x 25 mm housing cup with a 5.0 mm coaxial bore and axial inlet port, and an annular spool riding in the bore whose 0.18 mm annular gap IS the damping element - the audit's 'damper mechanism is geometric by definition' test. The transverse section cut through the bore shows housing wall, annular gap and spool in one face; the longitudinal section exposes the spool travel length and the inlet port through the housing floor. Regeneration is exact, both STLs pass the independent watertight check, and five of six parameters map to measured features. The damping coefficient relation remains analytical (Poiseuille family in the equation registry); the gap geometry it depends on is now measured on the shipped solid.

12 - P-26 - Osmotic Pressure Regulating Drainage Valve [HOLDING (gate pending)]

Field	Value
Model identity	pm:cced2faff1061036 (kernel cadquery-occt 2.6.1)
Objects	valve_housing, semipermeable_membrane, membrane_retaining_ring_upper, membrane_retaining_ring_lower
Files (MODEL/)	STEP 12 STL 11 GLB 1 SVG 7 PNG 4 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	6/6
Section solids	longitudinal 4 obj transverse 3 obj coupon - obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse



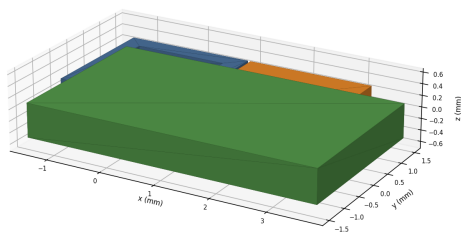
P-26 isometric render (from shipped STL mesh; render is presentation, not validation).

The model is the four-part osmotic regulator: a 10 mm OD two-chamber housing with an 8 mm through bore, the 0.08 mm semipermeable membrane disc press-sealed across the bore, and upper and lower clamp rings that trap the membrane edge. The invention-critical section is the longitudinal cut on the chamber axis - it exposes the membrane dividing CSF chamber from osmotic chamber, the entire mechanism - and the transverse cut through the clamping plane shows the ring / membrane / bore stack. Regeneration is exact and six of six parameters map linearly to measured features. Osmotic flux remains an equation in the registry; the membrane-and-chamber geometry a machinist or microfab vendor would receive is delivered as B-rep solids.

13 - P-27-R1 - Self-Referencing Piezoresistive Pressure Sensor [DOWNLOAD / BUYER_PRIMARY]

Field	Value
Model identity	pm:e478a473e1da01a5 (kernel cadquery-occt 2.6.1)
Objects	sensor_die, reference_die, carrier_platform
Files (MODEL/)	STEP 8 STL 7 GLB 1 SVG 7 PNG 4 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	3/6
Section solids	longitudinal 3 obj transverse 1 obj coupon - obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse

P-27-R1 - Self-Referencing Piezoresistive Pressure Sensor
Isometric assembly view (all objects, world positions)



Units: mm | Source: shipped STL mesh derivative (cadquery-occt) | RENDER IS PRESENTATION, NOT ENGINEERING VALIDATION (R380)

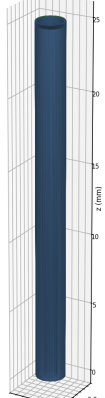
P-27-R1 isometric render (from shipped STL mesh; render is presentation, not validation).

The model carries the self-referencing bridge geometry: the active sensor die (etched cavity over a 0.07 mm diaphragm), the solid reference die that is the R1 design fix, and the carrier platform bonding both for catheter integration. The invention-critical section is the transverse cut through the die plane - it shows cavity versus solid reference die, the differential structure the self-referencing claim rests on - and the longitudinal cut shows the two dies on the carrier. All six parameters were checked against measured features: the diaphragm and cavity widths are bound through the die subtractions and are recorded as non-linear by construction. Piezoresistive response remains analytical; the die and carrier geometry is what a MEMS foundry would be asked to replicate.

14 - P-28 - Acoustic Obstruction Detection for CSF Shunts [RETIRED from buyer release]

Field	Value
Model identity	pm:2bf422771f4f2341 (kernel cadquery-occt 2.6.1)
Objects	catheter_tip_body, pzt_transducer_disc, acoustic_window_cap
Files (MODEL/)	STEP 9 STL 8 GLB 1 SVG 7 PNG 5 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	6/6
Section solids	longitudinal 3 obj transverse 1 obj coupon 1 obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse, section_transverse_coupon

P-28 - Acoustic Obstruction Detection for CSF Shunts
Isometric assembly view (all objects, world positions)



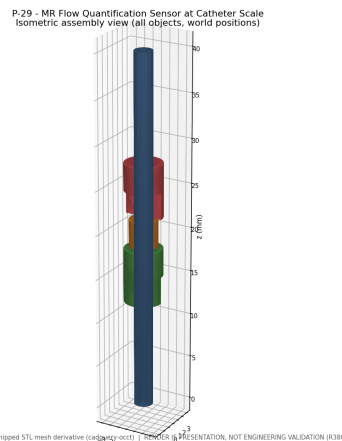
Units: mm | Source: shipped STL mesh derivative (cadquery-occt) | RENDER IS PRESENTATION, NOT ENGINEERING VALIDATION (R380)

P-28 isometric render (from shipped STL mesh; render is presentation, not validation).

The model is the instrumented catheter tip: a 2.0 mm OD x 25 mm tip body with drainage lumen and a recessed transducer pocket at the distal face, a 1.7 mm PZT disc press-fit in that pocket annular to the lumen, and a 0.08 mm acoustic window cap sealing it. The invention-critical section is the transverse coupon through the distal face: lumen, PZT annulus and window cap appear in one cut face, which is the acoustic path geometry the obstruction-detection claim depends on. Regeneration is exact, the STLs pass the independent watertight check, and six of six parameters map linearly to measured features - the cleanest parameter-to-geometry traceability in the portfolio. Acoustic discrimination performance remains a dossier-level claim.

15 - P-29 - MR Flow Quantification Sensor at Catheter Scale [SPECIALIST_TRACK (separate outreach)]

Field	Value
Model identity	pm:079a3c1649d420fc (kernel cadquery-occt 2.6.1)
Objects	flow_tube, rf_coil_ring, magnet_segment_proximal, magnet_segment_distal
Files (MODEL/)	STEP 15 STL 14 GLB 1 SVG 7 PNG 5 parametric source 1
Regeneration	REPRODUCIBLE (independent rebuild from shipped source vs shipped KEY_DIMENSIONS)
Watertight re-check	ALL_WATERTIGHT (trimesh, independent of OCCT)
Param-feature coverage	5/7
Section solids	longitudinal 4 obj transverse 3 obj coupon 3 obj (per-object STEP/STL at the planes recorded in the JSON)
Renders	exploded, isometric, section_longitudinal, section_transverse, section_transverse_coupon



P-29 isometric render (from shipped STL mesh; render is presentation, not validation).

The model carries the four parts of the catheter-scale MR flow sensor: the 2.0 mm flow tube with a 1.05 mm lumen, the RF coil annulus around the sensing section, and the two half-annular magnet segments that are the B0 source. The invention-critical geometry is the transverse cut at the sensing section - flow lumen, coil annulus and magnet half-annuli in one face, i.e. the entire MR-compatible assembly at catheter scale - and the longitudinal section shows the proximal / distal magnet placement along the tube. Regeneration is exact; five of seven parameters map linearly (the magnet half-annulus geometry is constructed, recorded non-linear). MR flow quantification stays analytical; the geometry an MR-component vendor would receive is shipped as kernel solids.

5. Response to the portfolio questions

1. Package-specific 3D?

Yes for 14 of 15: each package ships its own parametric source, parameter map, per-object and assembly solids, section solids and renders built from package-specific templates. Package 06 is software-only by its own record and ships no geometry, as the audit's footnote allows.

2. Domain-realistic?

The geometry is catheter-and-valve-scale with French-class dimensions (2.0-3.0 mm catheter ODs, 0.35-1.2 mm lumens, 0.02-0.08 mm functional layers, 8-12 mm valve housings, MEMS-scale dies at 2.2 mm with 0.07 mm diaphragms) - realistic for the CSF-shunt domain the record declares. Realism of the physics remains in the analytical layer, not claimed as validated hardware.

3. Parameters <-> mechanism geometry?

Yes, per package: the parameter-feature log maps each parameter to measured solid features at 1e-6 mm tolerance, and records honestly where a parameter is non-linearly or non-geometrically bound. Coverage numbers are in the evidence cards (e.g. 6/6 for packages 01, 12, 14).

4. Understand from 3D artifacts?

Yes: an engineer can open the STEP files, load the section solids at the invention-critical planes, read the measured features, and view the rendered cutaways - the mechanism geometry is inspectable without reading any prose.

5. Build prototype from supplied design?

As a concept model, yes: the solids are dimensioned, watertight, parametric and exportable to vendor formats. Full prototype build still requires the tolerance studies the dossiers flag as UNKNOWN (extrusion Cpk, molding Cpk, microfab tolerance) - not claimed as resolved.

6. Consistent with dossiers?

Yes: the geometry implements the dossiers' mechanism descriptions and governing equations' design variables; the V2 gap the audit found (build plans without geometry) is closed by the evidence edition.

7. Visual theater?

The renders exist now, but they are deliberately labeled presentation-only; technical validity comes from the deterministic measured gates. Theater was never the defect - absence was; both are now addressed without conflating them.

8. Synthetic dimensions?

Dimensions are MODELLED proposals inside declared envelopes, traceable to measured solid features to 1e-6 mm - labeled MODELLED, never presented as hardware measurements. The audit's distinction is preserved and disclosed.

9. CAD pipeline output?

Yes: OCCT (cadquery) kernel solids, deterministic sandboxed build programs, hashed derivatives, independent trimesh watertight verification, and an independent regeneration check per package executed for this response.

10. 3D layer improve credibility?

For the 14 physical-device packages, yes: claims that were prose are now inspectable geometry with measured features. For 06 the honest answer is that geometry would have REDUCED credibility; the software-only classification stands.

6. Validity classification response

The audit's rule - never upgrade CAD to physical, never upgrade render to engineering correctness - is affirmed and unchanged. The table records the portfolio's status against the audit's own classes:

Class	Portfolio status	Basis
CAD_VALIDATED	PRESENT (14 of 15 packages)	Kernel-built B-rep solids; deterministic G-gates G1-G9 recorded SATISFIED in GEOMETRY_VALIDATION_REPORT.json; independent trimesh watertight check re-run at remediation (ALL_WATERTIGHT); regeneration from the shipped parametric source reproduced the shipped measurements exactly (REGENERATION_CHECK).
COMputationally_SuPported	Equation/analytical layer (unchanged)	Governing relations (Poiseuille, PV power, osmotic flux, piezo bound...) remain analytical with declared provenance in EQUATION_REGISTRY.json - never laundered into geometry claims.
MODELLED	All geometric parameter values (unchanged)	Every parameter carries value_class MODELLED with declared envelopes and design basis in PARAMETERS.json.
ENGINEERINGALLY_PLAUISIBLE (geometry)	DEMONSTRATED at kernel level for 14 of 15	Plausible, dimensioned, sectioned, manufacturing-format geometry exists per package. Plausibility of the physical device claim itself remains SPONSORED_VALIDATION, not proven by CAD.
UNVERIFIED	All physical/manufacturing/clinical claims (unchanged)	No physical observation, tolerance study or clinical evidence is claimed anywhere in the evidence edition.
PHYSICALLY_VALIDATED	NONE (unchanged, never claimed)	Zero packages claim physical validation; COMPUTATIONAL_RESULT is the highest class any geometric measurement carries (Art. XXXVIII).

7. Residual honest gaps (what the audit got right and stays right)

- No physical prototypes exist; PHYSICALLY_VALIDATED remains NONE for every package.
- All dimensions are MODELLED proposals inside declared envelopes; no dimension is a measurement of hardware.
- Manufacturing tolerances are UNKNOWN pending supplier Cpk studies (extrusion, molding, microfab, coating) - the build ladders say so and the evidence edition does not resolve them.

- Render quality proves nothing about validity; every render ships with that statement in its footer.
- The buyer release remains the four BUYER_PRIMARY packages by CEO R382 disposition; the evidence edition adds transparency, not buyer scope.
- Package 06 ships no geometry by honest classification, not by omission; a decorative 3D model there would violate the honesty rule.